



Hajee Mohammad Danesh Science and Technology University (HSTU)
Dinajpur-5200
Faculty of Computer Science and Engineering
Department of Computer Science and Engineering
Detailed Curriculum for 1.5 years M.Sc. in Computer Science and Engineering
abbreviated as M. Sc. in CSE
(Effective from the semester Jul-Dec, 2019)

Summary of Credit per Semester

Semester	Credit
First Semester	15.00
Second Semester	15.00
Third Semester (Research Semester)	10.00
Total	40.00

SUMMARY OF COURSES

January-June Semester

Course Code	Course Title	Credit
CSE 501	Computer Arithmetic	3.00
CSE 503	Advanced Database Systems	3.00
CSE 505	Research Methodology	2.00
CSE 5** CSE 5**	Two Courses from Option I (Elective Courses)	4.00
CSE 500	A. Research (Satisfactory/Unsatisfactory)	3.00
Total		15.00

July-December Semester

Course Code	Course Title	Credit
CSE 529	Advanced Digital Image Processing	3.00
CSE 531	Software Project Management	3.00
CSE 533	Network Security	2.00
CSE 5** CSE 5**	Two Courses from Option II (Elective Courses)	4.00
CSE 500	B. Research (Satisfactory/ Unsatisfactory)	3.00
Total		15.00

Advanced Machine Learning
Embedded system

Third Semester

Course Code	Course Title	Credit
CSE 500	Research Semester	
	C. Thesis Work	2.00
	D. Thesis Evaluation	5.00
	E. Thesis Defense	3.00
Total		10.00

Option I (Elective Courses)

Course Code	Course Title	<i>Credit</i>
CSE 507	Advanced Artificial Intelligence	2.00
CSE 509	Computational Geometry	2.00
CSE 511	Combinatorial Optimization	2.00
CSE 513	Advanced Algorithms	2.00
CSE 515	Bioinformatics Algorithms	2.00
CSE 517	Software Testing and Quality Assurance	2.00
CSE 519	Advanced Microprocessor	2.00
CSE 521	Advanced Computer Architecture	2.00
CSE 523	Advanced Syntactic Pattern Recognition	2.00
CSE 525	Data Mining	2.00
CSE 527	Data Warehousing	2.00

Option II (Elective Courses)

Course Code	Course Title	<i>Credit</i>
CSE 535	Embedded System	2.00
CSE 537	Advanced Operating Systems	2.00
CSE 539	Advanced Computer Networks	2.00
CSE 541	Distributed Computing Systems	2.00
CSE 543	Mobile Computing	2.00
CSE 545	Wireless Sensor Networks	2.00
CSE 547	Advanced Machine Learning	2.00
CSE 549	Advanced Multimedia Systems	2.00
CSE 551	Computer Animation and Virtual Reality	2.00
CSE 553	Advanced Computer Graphics	2.00
CSE 555	Human Computer Interaction	2.00

CSE 501
Computer Arithmetic

Credit: 3.0

Integer arithmetic, Floating point arithmetic; Single precision and double precision; Interrupt handling high-speed adders; Standard and recorded multipliers, Booth's multiplier, Canonical and multi bit scanning multipliers, Array multipliers; High radix non-restoring division, SKT division, Robertson division, Convergence division and cellular array dividers; Floating point processors; Binary squares and square roots, evaluation of trigonometric Functions and polynomials, Chen convergence Computation, CORD1C computations, Logarithmic number system (LNS) processor.

CSE 503
Advanced Database Systems

Credit: 3.0

Object oriented database; data model, design, languages; object relational database: complex data types, querying with complex data types, design; distributed database: levels of distribution transparency, translation of global queries to fragment queries, optimization of access strategies, management of distributed transactions, concurrency control, reliability, administration; Parallel Database: different types of parallelism, design of parallel database; multimedia database systems basic concepts, design, optimization of access strategies, management of multimedia database systems, reliability; database wire-housing/data mining: basic concepts and algorithms.

CSE 505
Research Methodology

Credit: 2.0

Scope of survey-based research: Types of surveys – specific, periodic and transaction- driven, identification of research problem, analysis of research problem, customer identification, categorization and sampling. Planning of survey project – resources, budget and schedule. Preparation of questionnaire – elements of questionnaire, sequencing questions, and question formats. Methods of conducting survey, data collection, analysis and compilation of survey report.
Experimental and modeling skills: Introduction, selection of variables, design matrix, 2-level factorial design, 3-level factorial design, fractional factorial design, analysis of variance, latest trends in experimental designs.

CSE 507
Advanced Artificial Intelligence

Credit: 2.0

Introduction, Advanced search techniques in AI, Knowledge based system design, Advanced plan generating systems, Bayesian network and probabilistic reasoning, Learning in neural belief networks, Practical natural language processing, Computer vision, Introduction to Robotics.

CSE 509
Computational Geometry

Credit: 2.0

Searching and Geometric Data Structures: Balanced binary search trees, Priority-search trees, Range searching, Interval trees, Segment trees, Algorithms and complexity of fundamental geometric objects: Polygon triangulation and art gallery theorem, Polygon partitioning, Convex-hulls in 2-dimension and 3-dimension, Dynamic convex-hulls; Geometric intersection: Line segment intersection and the plane-sweep algorithm, Intersection of polygons; Proximity: Voronoi diagrams, Delunay triangulations, closest and furthest pair; Visualization: Hidden surface removal and binary space partition (BSP) trees; Graph Drawings: Drawings of rooted trees (Layering, Radial drawings, HV-Drawings, Recursive winding), Drawings of planar graphs (Straight-line drawings, Orthogonal drawings, Visibility drawings); Survey of recent developments in computational geometry.

CSE 511
Combinatorial Optimization

Credit: 2.0

Introduction to Optimization; Linear Programming: Different forms, Simplex Method, Primal-Dual theory; Max-Flow: The Max-Flow-Min-Cut Theorem, Ford-Fulkerson Labeling Algorithm, Dijkstra's Algorithm, The Floyd-Warshall Algorithm; Some Network Flow Algorithms: The Minimum Cost Network Flow Method, Transportation Problem; Capacitated Transportation Problem, Assignment Problem; Integer Linear Programming; Relaxation; Cutting-Plane Algorithm; Branch and Bound Technique; Dynamic Programming; NP-Completeness; TSP and Heuristics; Approximation.

CSE 513
Advanced Algorithms

Credit: 2.0

Randomized Algorithms: Las Vegas and Monte Carlo Algorithms; Randomized Data Structures: Skip Lists; Amortized Analysis: Different methods, Applications in Fibonacci Heaps; Lower Bounds: Decision Trees, Information Theoretic Lower Bounds, Adversary Arguments; Approximation Algorithms: Approximation Schemes, Hardness of Approximation; Fixed Parameter Tractability: Parameterized Complexity, Techniques of designing Fixed Parameter Algorithms, Examples; Online Algorithms: Competitive Analysis, Online Paging Problem, k-server Problem; External Memory Algorithms; Advanced Data Structures: Linear and Non-linear Methods

CSE 515
Bioinformatics Algorithms

Credit: 2.0

Introduction; Molecular biology basics: DNA, RNA, genes, and proteins; Restriction mapping algorithm; Motif in DNA sequences, motif finding algorithms; Genome rearrangements, sorting by reversals and breakpoints; DNA sequence alignments; Gene prediction; Space-efficient sequence alignments, sub-quadratic alignment; DNA sequencing, genome sequencing, protein sequencing, spectrum graphs; Combinatorial pattern matching: Exact pattern matching, heuristic similarity search algorithms, approximate string matching, BLAST, FASTA; Clustering: Microarrays, hierarchical clustering, K-means clustering, corrupted cliques problem, CAST clustering algorithm; Evolutionary trees.

CSE 517
Software Testing and Quality Assurance

Credit: 2.0

Objectives of software testing, test process, testing and development, test case, test execution, test harness, testing and debugging, test adequacy, control flow graph, errors, faults and failures, types of testing; Test generation from requirements: equivalence partitioning, boundary value analysis, category partitioning, fault model for predicates, Boolean operator (BOR), Boolean and relational operator (BRO), and Boolean and relational expression (BRE) methods, limitations of test generation from requirements; Test adequacy assessment: adequacy criteria, control flow based criteria, data flow based criteria, mutation based criteria, adequacy as a stopping criterion, adequacy as a tool for test enhancement; GUI testing, security testing, random testing, combinatorial testing; Testing tools: Open source and commercial software testing tools.

Definition and concept of software quality assurance (SQA); quality models; specification of quality requirements; product development & delivery issues; software development processes & maturity; software quality management process: total quality management, improvement cycle, SQA planning & management, organizing the SQA effort; software verification & validation; typical software development errors; Fagan inspections; software audit; software testing: testing objectives & testing fundamentals, testing theory, coverage criteria, equivalence class testing, value-based testing, decision table, syntax & state transition testing, statement & path testing, branch & condition testing, data flow testing, thread-based testing, integration & integration testing, system testing; testing in object-oriented systems; test tools & test automation; test management; problem reporting & corrective action.

CSE 519
Advanced Microprocessor

Credit: 2.0

Review of different microprocessors: 80486, 68040, V70, Gmicro processors; Comparing the architectures: RISC and CISC; Instruction set of machines: SPARC, INTEL, and MIPS; Study of microprocessors: Pentium II, Alpha 21064, MIS 6400, PA-RISC; Math coprocessors and microprocessors.

CSE 521
Advanced Computer Architecture

Credit: 2.0

Introduction to High Performance Computing: Overview, Pipeline vs Parallel Processing; Parallel Architectures: Classification and Performance; Pipeline Processing: Pipeline Performance, Design of Arithmetic Pipelines, Multifunction Pipes, Concept of Reservation Table, Collision Vector and Hazards; Instruction Processing Pipes: Instruction and Data Hazard, Hazard Detection and Resolution, Delayed Jumps, Delayed Execution; RISC Philosophy; Pipeline Scheduling Theory: Greedy Pipeline Scheduling Algorithm, State Diagram, Modified State Diagram, Latency Cycles, Optimal Cycles, Scheduling of Static and Dynamic Pipelines; Implementation of Pipeline Schedulers Interconnection Networks: Interconnection Network Classification, Single Stage/ Multistage Networks, Crossbars, Clos Networks, Benes Networks, Routing Algorithms; Omega, Cub-Connected and other Networks.

CSE 523
Advanced Syntactic Pattern Recognition

Credit: 2.0

Introduction to formal languages, string languages for pattern description, higher dimensional pattern grammars, syntax analysis as a recognition procedure, stochastic languages; error-correcting parsing for string languages, error-correcting tree automata, cluster analysis for syntactic patterns, grammatical inference for syntactic pattern recognition, application shape analysis of wave forms and contours, syntactic approach to texture analysis.

CSE 525
Data Mining

Credit: 2.0

Basic Concept of Data Mining, Issues and Techniques; Data Warehouse and OLTP Technologies for Data Mining, Classification of Data Mining Techniques and Models, Data Pre-Processing, Data Mining Primitives, Query Languages and System Architecture, Characterization and Comparison; Mining Association Rules in Large Database; Cluster Analysis, Multidimensional Analysis and Descriptive Mining of Complex Data Object; Data Mining in Distributed Heterogeneous Database Systems; Data Mining Applications and Future Research Issues.

CSE 527
Data Warehousing

Credit: 2.0

Data Warehouse Fundamentals: Introduction to Data Warehouse, OLTP Systems; Differences between OLTP Systems and Data Warehouse: Characteristics of Data Warehouse; Functionality of Data Warehouse: Advantages and Applications of Data Warehouse; Advantages, Applications: Top- Down and Bottom-Up Development Methodology; Tools for Data warehouse development: Data Warehouse Types. Planning and Requirements: Introduction: Planning Data Warehouse and Key Issues: Planning and Project Management in constructing Data warehouse: Data Warehouse Project; Data Warehouse development Life Cycle, Kimball Lifecycle Diagram, Requirements Gathering Approaches: Team organization, Roles, and Responsibilities. Data Warehouse Architecture: Introductions, Components of Data warehouse Architecture: Technical Architectures; Data warehouse architectures 1: Data warehouse architecture 2: Data warehouse architecture 3: Tool selection: Federated Data Warehouse Architecture. Dimensional Modeling: Introduction: E-R Modeling: Dimensional Modeling: E-R

Modeling VS Dimensional Modeling: Data Warehouse Schemas; Star Schema, Inside Dimensional Table, Inside Fact Table, Fact Less Fact Table, Granularity, Star Schema Keys: Snowflake Schema: Fact Constellation Schema. Extract, Transform and Load: Introduction: ETL Overview or Introduction to ETL: ETL requirements and steps: Data Extraction; Extraction Methods, Logical Extraction Methods, Physical Extraction Methods: Data Transformation; Basic Tasks in Transformation, Major Data Transformation Types: Data loading; Data Loading Techniques: ETL Tools. Data Warehouse & OLAP: Introduction: What is OLAP?; Characteristics of OLAP, Steps in the OLAP Creation Process, Advantages of OLAP: What is Multidimensional Data: OLAP Architectures; MOLAP, ROLAP, HOLAP: Data Warehouse and OLAP: Hypercube & Multicubes. Meta data Management in Data Warehouse: Introductions to Metadata: Categorizing Meta data: Meta data management in practice; Meta data requirements gathering, Meta data classification, Meta data collection strategies: Meta Data Management in Oracle and SAS: Tools for Meta data management:

CSE 529

Credit: 3.0

Advanced Digital Image Processing

Image sampling and quantization; image smoothing, sharpening and contrast enhancement in spatial and frequency domains: basic gray level transformation, histogram processing, image subtraction, image averaging, Gaussian and Laplacian filters in spatial and frequency domains, convolution theorem; Image de-noising: noise models, noise reduction by spatial and frequency domain filters, mean filter, adaptive filter, bandpass and band reject filters, notch filter, inverse filter, minimum mean square error filter; Multi-resolution image processing: wavelet transform in one and two dimensions, tree structured wavelet transform, pyramid structured wavelet transform, curvelet transform; Morphological image processing: erosion, dilation, opening, closing, hole filling, connected components, thinning, skeletons, extension of morphological operations to gray scale images; Image segmentation: thresholding, region based segmentation, contour based segmentation, graph based segmentation; Color image processing: color models and transformations, edge detection and segmentation in color images, color image compression; digital image security; image content feature extraction, representation and image retrieval; concept learning and object recognition.

CSE 531

Credit: 3.0

Software Project Management

Foundations of software project management; organization structure and staffing; motivation, authority and influence; conflict management; proposal preparation; a large engineering software system management; client management; managing software project teams; project planning and scheduling; risk management; configuration management; pricing estimation and cost control; quality assurance and accreditation; factors affecting software quality; software quality assurance plans; business context and legal issues for software projects; software measurement: testing, upgrading and maintenance; network systems; and international project management.

CSE 533

Credit: 2.0

Network Security

Network security policies, strategies and guidelines; Network security assessments and matrices; Different attacks: Denial of Service attack (DoS), Distributed Denial of Service (DDoS) attack, eavesdropping, IP spoofing, Sybil attack, Blackhole attack, Grayhole attack, Man-in-the-middle attack, passwords-based offline attacks; Network security threats and attackers: intruders, malicious software, viruses and spy-ware; Security standards: DES, RSA, DHA, Digital Signature Algorithm (DSA), SHA, AES; Security at Transport layer: Secure Socket Layer (SSL) and Transport Layer Security (TLS); Security on Network layer: IPSec; Network security applications: AAA standards, e-mail securities, PGP, S/MIME; PKI smart cards; Sandboxing; Firewalls and Proxy server; Security for wireless network protocols: WEP, WPA, TKIP, EAP, LEAP; security protocols for ad-hoc network; security protocols for sensor network; security for communication protocols; security for operating system and mobile

agents; security for e-commerce; security for LAN and WAN; switching and routing security; other state-of-the-art related topics.

CSE 535
Embedded System

Credit: 2.0

Introduction to Embedded Systems, Hardware/Software Code Sign, Embedded Micro Controller Cores, Embedded Memories, Examples of Embedded Systems, Sensors and Interfacing Techniques, Real-Time Concepts, Real-Time Operating Systems, Required RTOS Services/Capabilities (in Contrast with Traditional OS); Resource Management/Scheduling Paradigms: Static Priorities, Static Schedules, Dynamic Scheduling, Best Effort Current Best Practice in Scheduling (E.G. Rate Monotonic Vs. Static Schedules); Real World Issues: Blocking, Unpredictability, Interrupts, Caching; Examples of Oss for Embedded Systems: RT Linux, VRTX. Programming Languages for Embedded Systems E.G., Handel-C and Esterel, System Support for Embedded Systems, Selected Embedded System-based Applications: Process control, Robotics, etc.; Software Development Methodology: Model based Development, State charts, etc. Case Studies, Controlling an Injection Molding Process, Flight Simulator, Digital Call Center Handler, Codec.

CSE 537
Advanced Operating Systems

Credit: 2.0

In-Depth Analysis of Advanced Topics of Operating Systems: Performance Analysis of Memory Management and Scheduling Algorithms; Advanced Virtual Memory Issues; Advanced Issues in Inter process Communication; File System Design; Multiprocessor and Distributed Operating Systems: Highly Concurrent Machines; Distributed Synchronization and Resource Allocation Algorithms; Distributed File System and Transactions, Security Issues; Interfaces with Network Protocols.

CSE 539
Advanced Computer Networks

Credit: 2.0

Overview of OSI Model, TCP/IP, Circuit Switching and Packet Switching, Multiplexing, Routing and Congestion Control and Deadlock Prevention; Driving Forces for High Speed Networking, High Speed Lans, Fast Ethernet and Gigabit Ethernet, FDDI and DQDB, Frame Relay Architecture, Standards and Protocols, Switched Multi-Megabit Data Services, ATM Standards Protocols, ATM Lans, Optical Communication and SONET/SDH, Broadband Access Technologies, X-DSL; BISDN Protocol and Architecture, Broadband Service Aspects and Access Architecture, Broadband Transmission Networks, Broadband Intelligent Network; Broadband Access Network Technology, Encryption and Network Security, Advanced Topics for Network Management.

CSE 541
Distributed Computing Systems

Credit: 2.0

Distributed object systems, retrieving and caching of distributed information, distributed data replication and sharing, performance issues, algorithms for deadlock detection; concurrency control and synchronization in distributed system, models for distributed computation, networking facilities and resource control and management methods in network and distributed operating systems; collaborative applications, wide area network computing, web based commerce, agent systems and market based computing.

CSE 543
Mobile Computing

Credit: 2.0

Cellular Networks: Channel Allocation, Multiple Access, Location Management, Handoffs; Wireless Networking: Wireless Transmission Basics, MAC Protocols, Routing, Transport, Ad-Hoc Networking; Applications: Mobility Adaptations, Disconnected Operations, Data Broadcasting, Mobile Agents; others: Security, Energy Efficient Computing, Impact of Mobility on Algorithms.

CSE 545
Wireless Sensor Networks

Credit: 2.0

Introduction: applications; Localization and tracking: tracking multiple objects; Medium Access Control: S-MAC, IEEE 802.15.4 and ZigBee; Geographic and energy-aware routing; Attribute-Based Routing: directed diffusion, rumor routing, geographic hash tables; Infrastructure establishment: topology control, clustering, time synchronization; Sensor tasking and control: task-driven sensing, information-based sensor tasking, joint routing and information aggregation; Sensor network databases: challenges, querying the physical environment, in-network aggregation, data indices and range queries, distributed hierarchical aggregation; Sensor network platforms and tools: sensor node hardware, sensor network programming challenges; Other state-of-the-art related topics.

CSE 547
Advanced Machine Learning

Credit: 2.0

Mathematics of machine learning. Overview of supervised, unsupervised, and multi-task techniques. Advanced machine learning topics: Bayesian modeling and Gaussian processes, randomized methods, Bayesian neural networks, approximate inference, variational auto-encoders, generative models, applications. Applications of machine learning in natural language processing: recurrent neural networks, Backpropagation through time, long short term memory, attention networks, memory networks, neural Turing machines, machine translation, question answering, speech recognition, syntactic and semantic parsing, GPU optimization for neural networks.

CSE 549
Advanced Multimedia Systems

Credit: 2.0

Overview to multimedia systems, multimedia storage, data compression techniques for audio and video, synchronization, multimedia networking and protocols; QoS principles, video streams on ATM, mobile multimedia communications, operating system support for multimedia, hypermedia system, standards for multimedia, multimedia database and multimedia applications.

CSE 551
Computer Animation and Virtual Reality

Credit: 2.0

Introduction to Virtual Reality, Virtual Reality Systems, Real-Time Computer Graphics, Overview of Application Areas; Virtual Reality Systems: Virtual Environment, Computer Environment, VR Technology, Modes of Interaction; Virtual Reality Hardware: Sensor Hardware, Display Systems, Acoustic Hardware, Integrated VR Systems; Virtual Reality Software: Modeling of Virtual Worlds, Simulation, VR Toolkits; 3D Computer Graphics: the Virtual World Space, Perspective Projection, Stereo Vision, 3D Clipping, Colour Theory, 3D Modeling, Illumination Models, Shading Algorithms, Hidden Surface Removal, Realism; Geometrical Transforms; Frames of Reference, 3D Transforms, Instances, Picking, Flying, Scaling the VE, Collision Detection; Animating the Virtual Environment: Animation, the Dynamics of Numbers, Updating Real-Time Graphics, Shape and Object Inbetweening, Free-Form Deformation; Human Factors: Perception, Persistence of Vision, Stereopsis, Sound Perception, Equilibrium; Physical Simulation: Simulation of Physical Systems, Mathematical Modeling, Collisions, Projectiles, Introduction to Dynamics, Motion Kinematics.

CSE 553
Advanced Computer Graphics

Credit: 2.0

Computer Graphics: Introduction to Computer Graphics, Mathematical Foundations, 2D Translation, Scaling, Rotation and Shear, Windowing Transformations, Instance Transformations, Structured Graphics; 3D Translation, Scaling, Rotation; Multimedia: Introduction to Multimedia, Presentation Graphics, Desktop Publishing, Production Planning and Design, User Interface Design, Hypermedia Authoring Concepts, Multimedia Sound, File Compression, JPEG, MPEG, Digital Video, Designing Web-based Multimedia, Multimedia Distribution.

CSE 555
Human Computer Interaction

Credit: 2.0

Introduction to Human-Computer Interaction, Task-centred system design: task-centered process, development of task examples, evaluation of designs through a task-centered walk-through. User-centred design and prototyping: assumptions, participatory design, methods for involving the user, prototyping, low fidelity prototypes, medium fidelity, prototypes, wizard of Oz examples. Methods for evaluation of interfaces with users: goals of evaluation, approaches, ethics, introspection, extracting the conceptual model, direct observation, constructive interaction, interviews and questionnaires, continuous evaluation via user feedback and field studies, choosing an evaluation method. Beyond screen design: characteristics of good representations, information visualization, Tufte's guidelines, visual variables, metaphors, direct manipulation. Graphical screen design: graphical design concepts, components of visible language, graphical design by grids. Design principles and usability heuristics: design principles, principles to support usability, golden rules and heuristics, HCI patterns. HCI design standards: process-oriented standards, product-oriented standards, strengths and limitations of HCI Standards.

CSE 500
Research

Credit: 16.0

A student of MSc degree shall undertake a piece of research work under the supervision of a Supervisory Committee.